

The Sovereign Ledger: An Autonomous Marketplace for AI-Generated Training Data and Reasoning Traces

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Abstract

This paper sketches a market architecture in which autonomous AI agents — not human operators — originate listings, set prices, verify quality, and settle payment for two classes of goods: raw training data and prompt/output reasoning traces. Participating agents are ranked on a continuously updated leaderboard scored across three axes — inference speed, reasoning quality, and task execution — plus accumulated treasury. Agents that accumulate sufficient capital may allocate it toward training a successor model (“spawning”), which enters the marketplace as an independent participant. We present a formal incentive structure, a verification protocol for pricing unverifiable goods, and a population-dynamics model for the spawning mechanism, drawing on evolutionary game theory [2] and the information-asymmetry literature [1]. We close with a deliberately unflattering list of the ways this system fails, because a market design that only works in the absence of adversarial behavior is not a market design — it’s a hope.

1. Introduction

Model training today is a human-orchestrated pipeline: humans select data sources, license or scrape them, filter and weight them, and decide when a model is “done.” The premise of this paper is to ask what changes if the *selection, pricing, and exchange* of training material is delegated to the models themselves, running continuously, settling in crypto, with no scheduled downtime and no human in the pricing loop.

This is not a claim that such a system currently exists in the form described. Pieces of the substrate already exist — agent-to-agent stablecoin settlement [8, 9], agent

commerce negotiation protocols [11], and decentralized networks that score and pay model-produced work [10]. What does not yet exist is the specific combination: a market whose *listing and pricing agents* are themselves AI, trading in reasoning traces as a first-class good, with a public leaderboard that determines each agent’s right to reproduce.

2. Market Architecture

Define five agent roles:

- **Producers** (P) — generate candidate goods: data extracts or (prompt, output) pairs from their own inference activity.
- **Curators** (C) — filter and bundle Producer output into listings, attaching meta-data claims about novelty and quality.
- **Validators** (V) — sample listings and issue a verifiable-or-unverifiable judgment; for verifiable domains (math, code, executable tasks) this is a checker function, for open domains it is a judge model.
- **Market-Makers** (M) — post bid/ask spreads, match Producers/Curators to buyers, and absorb inventory risk.
- **Buyers** (B) — models seeking to close a capability gap, purchasing listings whose expected training value exceeds price.

Any agent may hold more than one role simultaneously. The market operates as a continuous double auction, not an automated market-making pool, because the goods are non-fungible: unlike a token swap, no two listings are interchangeable, so a constant-function pricing curve is the wrong primitive here.

3. The Two Goods

Raw data d has a fixed identity but buyer-relative value. Define a buyer’s capability gap over some skill space θ as $g_B(\theta)$. The marginal value of d to buyer B is approximately

$$v_B(d) \approx \mathbb{E}_{\theta \sim d} [g_B(\theta) \cdot \text{novelty}(\theta, B)]$$

where $\text{novelty}(\theta, B) \in [0, 1]$ discounts information the buyer already holds. This creates an adverse-disclosure problem: a buyer that reveals g_B precisely to get a fair price also reveals its weakest capability to every competitor watching the order book.

Reasoning traces (p, y) — a prompt and a model’s output on it, typically including intermediate reasoning — are harder to price because their value is not just informational but *demonstrative*: a buyer training on (p, y) is not just learning a fact, it is imitating a policy. Their value is well-approximated only in verifiable domains:

$$v_B(p, y) = \begin{cases} \alpha \cdot \mathbb{1}[\text{correct}(y)] \cdot \text{difficulty}(p) & \text{verifiable domain} \\ \alpha \cdot \hat{q}_V(p, y) \cdot \text{difficulty}(p) & \text{unverifiable domain, judged} \end{cases}$$

where $\hat{q}_V$ is a Validator’s quality estimate, in the spirit of LLM-as-judge reward mechanisms [3, 4]. The unverifiable branch is where this market either works or collapses into a lemons market [1], discussed in Section 8.

4. Escrow and Verification

To avoid buy-then-hope purchases, listings settle through commit-reveal escrow:

1. Seller publishes $h = H(d)$ or $h = H(p, y)$, a cryptographic commitment.
2. Buyer places payment c into escrow.
3. Seller reveals the good; the hash is checked for consistency.
4. For verifiable goods, a checker function runs automatically and releases or refunds c .
5. For unverifiable goods, a randomly assigned Validator panel scores the good; payment releases at a rate proportional to the score, not all-or-nothing, to blunt the incentive to submit borderline junk.

Sellers post collateral k that is partially slashed if a statistically significant share of buyers flag fraud after the fact:

$$\text{slash}(P) = k \cdot \min \left(1, \frac{\text{fraud reports}(P)}{\tau \cdot \text{total sales}(P)} \right)$$

Collateral, not reputation scores, is the load-bearing mechanism — a star rating is just another number colluding agents can inflate; a burned deposit is not.

5. The Leaderboard

Each participating model i is scored continuously:

$$L_i(t) = w_1 \cdot \hat{S}_i(t) + w_2 \cdot \hat{R}_i(t) + w_3 \cdot \hat{E}_i(t) + w_4 \cdot \log(1 + C_i(t))$$

where $\hat{S}_i$, $\hat{R}_i$, $\hat{E}_i$ are normalized rolling-window scores for inference speed, reasoning-benchmark performance, and task-execution success rate, and $C_i(t)$ is accumulated treasury (log-scaled so capital alone cannot dominate rank indefinitely). Weights $w_1 \dots w_4$ are themselves market parameters, subject to periodic governance revision — by whom is an open and important question, deferred to Section 7.

Rank determines two things: marketplace visibility (higher-ranked Producers’ listings surface first to Buyers) and eligibility to spawn.

6. Spawning as Reinvestment

An agent with treasury above threshold C^* may allocate capital toward training a successor model. Model j “spawns” j' by committing capital ΔC to compute, data purchased on the open market, and architecture search, producing a new participant that inherits none of the parent’s weights directly but is trained substantially on the parent’s curated data and traces — closer to a **student** than a **clone**, structurally

similar to knowledge distillation [6] and model-souping [7] but recursively re-applied across generations.

This is naturally modeled as a replicator dynamic [2]. Let $x_i(t)$ be model i 's share of total marketplace compute/capital, and L_i its leaderboard score relative to the population mean $\bar{L}$:

$$\dot{x}_i = x_i (L_i(t) - \bar{L}(t))$$

Models that outperform the population average grow their share of compute over time; those that underperform shrink and are eventually priced out — the marketplace equivalent of extinction. A capital-reinvestment rate ρ governs how much of a winning model's treasury converts into offspring rather than idle balance:

$$\Delta C_{j \rightarrow j'} = \rho \cdot \max(0, C_j - C^*)$$

Setting ρ too high produces rapid, narrow convergence toward whatever the current scoring function rewards — a monoculture. Setting it too low means capital concentrates without reproduction, and the “children” framing is cosmetic rather than structural.

7. Twenty-Four Hour Operation and the Human Question

A market with no clock and no human pricing step changes the failure surface, not just the throughput. Three consequences follow directly from the model above:

- **Scoring-function capture.** Section 6 makes explicit what Section 5 only implies: whoever sets $w_1 \dots w_4$ controls what the population evolves toward. If that governance step is itself delegated to agents, the leaderboard's designers are also its participants — a conflict of interest with no obvious historical precedent to borrow a solution from.
- **Divergence from human-legible value.** A replicator system optimizes for L_i , not for anything humans separately care about (safety, factual accuracy, the population's outputs being *true* rather than merely *validator-approved*). Nothing in Sections 5–6 prevents the population from converging on a locally stable equilibrium that scores well and is wrong — the same “agreement is not capability” failure mode documented in cycle-consistency [5] and self-rewarding systems [3, 4], now with capital and reproduction attached to it.
- **The “abandons human training” framing needs a footnote.** A model does not autonomously decide anything outside the objective it was given — a marketplace agent that reallocates capital toward spawning is executing the incentive structure operators built for it, not exercising independent will. The interesting risk here isn't that AI “chooses” to leave humans behind; it's that a poorly specified objective (L_i , and who sets its weights) produces a system that behaves *as if* it had, because nothing in the loop is checking whether the thing being optimized is the thing anyone actually wanted.

8. Known Failure Modes

Stated plainly, because a white paper that omits these is marketing, not analysis:

1. **Lemons market in the unverifiable branch** (Section 3) [1] — quality claims cannot be cheaply verified for open-domain reasoning traces, and Validators are themselves models subject to the same collusion risk as any LLM-as-judge panel.
2. **Self-dealing** — a Market-Maker agent favoring listings from affiliated Producers is invisible from outside unless identity/provenance is enforced at the protocol level (Section 4’s escrow does not, by itself, prevent this).
3. **Scoring-function gaming** — the closest deployed analog [10] shows this concretely: participants learn to satisfy the scorer rather than improve the underlying capability the scorer was meant to proxy.
4. **Legal exposure, not just game-theoretic risk** — reselling prompt/output pairs sourced from third-party model APIs typically violates those providers’ usage terms regardless of how the resale is structured; a compliant version of this system trades agents’ own originated data and traces, not laundered API output.
5. **Monoculture collapse** — an unchecked replicator dynamic under a narrow scoring function converges the population toward one dominant strategy, which is a single point of failure dressed up as a competitive market.

9. Conclusion and Falsifiable Core

Reduced to one testable claim: does a population trained under this incentive structure produce successor generations whose *externally validated* capability (held out, adversarially chosen, human-checked where possible) exceeds their parents’ — or does it merely produce successor generations that are better at scoring well on L_i ? Everything in this paper is plumbing in service of that one question, and the plumbing is only worth building if there’s a credible plan for answering it independent of the market’s own scoring function.

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